	Faculty Department: CSE **Academic Year: 2025-2026**	**Submission Date: 10.02.2026**
	B.E: CS **Specialisation: NA**	
	Student Department: CSE **Semester: IV** **Assignment No: I**	
	Sub. Code & Name : CS22403 & Design and Analysis of Algorithms	**Max. Marks: 50**

Q. No.	Question / Task / Activity	CO*	RBT Level#	Marks
1	(i) **Solve below function:** If *T1(n) = O(f(n))* and *T2(n) = O(g(n))*, then show that: (a) T1(n) + T2(n) = max (O(f(n)), O(g(n))) (b) T1(n) * T2(n) = O(f(n) * g(n)) (ii) **Derive the mathematical analysis to compute the efficiency on:** (a) Finding Factorial of ‘ N’ integers (b) To count binary digits (c) To Find Unique element	1	U	10
2	A student record management system stores student roll numbers in an array. At the beginning of each semester, new students are admitted one by one, and their roll numbers must be inserted into the correct position so that the list remains sorted at all times. The system uses the Insertion Sort algorithm to maintain the sorted order of roll numbers. i. Explain how Insertion Sort operates in the given scenario. ii. Determine the best-case, average-case, and worst-case time complexity of Insertion Sort. iii. Analyze the efficiency of Insertion Sort changes as the number of students increases. iv. Justify whether Insertion Sort is suitable for this application.	1	AN	10
3	**Apply Masters Theorem** (a) $T(n) = 3T(n/2) + n^2$ (b) $T(n) = 4T(n/2) + n^2$ (c) $T(n) = T(n/2) + 2^n$ (d) $T(n) = 16T(n/4) + n$	2	AP	10
4	**Sorting Using Moving Intervals:** There are C cakes in a row, numbered from 1 to C. There are N children, each of whom have selected a consecutive set of cakes to eat. That is, Child i has decided to eat all the cakes from Si to Ei, end points inclusive. If there is a cake which appears in some two children`s set, then they will fight because both of them want to eat that cake, and you don't want that to happen.	2	E	30

You will be given an integer K which will be either 0 or 1. If K is 0, then you should find out if some two children will fight. Print "Good" if no one fights, and "Bad" if someone fights.

If K is 1, then you can persuade at most one child to change his decision to some other set of cakes. But the number of cakes that he eats must be the same. That is, if Child i had initially decided that he wants to eat the cakes from Si to Ei, then you could persuade the child to instead eat the cakes from X to Y instead, for any valid X and Y (ie. $1 \leq X \leq Y \leq C$), provided that the number of cakes is the same (ie. $Ei - Si + 1 = Y - X + 1$). If after persuading at most 1 Child to change his decision, no fights happen, then print "Good". But if no matter what you do, someone will fight, then print "Bad".

Explanation

Sample 1: Child 1 wants to eat the second cake, and Child 2 wants to eat Cakes 3, 4 and 5. So there is no fight, and the answer is "Good".

Sample 2: Child 1 wants to eat Cake 2 and Cake 3, and Child 2 wants to eat Cakes 2, 3, 4 and 5. Both of them want to eat Cakes 2 and 3, and hence it could lead to a fight. And because K = 1, we can persuade one of the children to change their decision. For instance, we could persuade Child 1 to change his decision from [2, 3] to [1, 2]. But even after this, both of them want to eat Cake 2. You can verify that no matter how we persuade at most 1 child, they will end up fighting. Hence the answer is "Bad".

i. "Compile and run" to evaluate your submission against the public test cases your own language (C/C++/Java/Python3)
ii. Prove your solution for 3 test case give below
iii. Solution of Q4 can be submitted as **hard copy print out** along with above Q1,Q2,Q3 questions hand written assignment.

Test cases:

```
5 2 0
2 2
3 5
Answer: Good
5 2 1
2 3
2 5
Answer :  Bad
```

Source: NPTEL Assignment
https://onlinecourses.nptel.ac.in/noc26_cs67/progassignment?name=255

			Total	**50**